

STANDARD OPERATING PROCEDURE

WC-100 PRODUCT QUALITY REVIEW AND RELEASE

Document Number: SOP-WC-100-001

Revision: A

Document Status: Demonstration Document

Document Type: Standard Operating Procedure

Product: WC-100 Industrial Wear Component

Process: Product Quality Review and Release

Prepared By: Technical Documentation

Effective Date: 30 August 2026

Review Frequency: Annual or when significant process changes occur

DOCUMENT CONTROL

Field	Details
Document Number	SOP-WC-100-001
Document Title	WC-100 Product Quality Review and Release
Revision	A
Document Type	Standard Operating Procedure
Product	WC-100 Industrial Wear Component
Process Owner	Quality
Primary Users	Quality, Manufacturing, Engineering and Operations
Status	Demonstration
Effective Date	30 August 2026
Review Frequency	Annual or upon significant process change

1. PURPOSE

This Standard Operating Procedure defines the controlled process for reviewing, verifying and releasing completed WC-100 Industrial Wear Components for the next stage of the product lifecycle.

The purpose of this procedure is to ensure that product release decisions are based on complete and accurate manufacturing, inspection and quality information.

This procedure establishes a consistent process for:

- Reviewing manufacturing records
- Confirming required inspections are complete
- Verifying product conformity
- Checking traceability
- Reviewing non-conformances
- Confirming required approvals
- Recording the release decision
- Maintaining supporting records

The procedure supports consistent quality decisions, traceability and operational control.

2. SCOPE

This SOP applies to WC-100 components manufactured through the applicable production process and submitted for final quality review.

The procedure covers activities from completion of manufacturing through final quality disposition.

It applies to personnel involved in:

- Manufacturing
- Quality
- Engineering
- Production supervision
- Document control
- Product release

This SOP does not replace detailed manufacturing instructions, engineering specifications or inspection work instructions.

The following documents provide detailed requirements for specific activities:

- WC-100 Product Specification
 - WC-100 Manufacturing Work Instruction
 - WC-100 Inspection Plan
 - Applicable Quality Procedures
 - Applicable Engineering Documentation
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3. PROCESS OVERVIEW

The WC-100 product release process follows the sequence below:

Manufacturing Completion



Record Submission



Documentation Review



Inspection Verification



Traceability Verification



Non-Conformance Review



Engineering / Quality Review, Where Required



Final Release Decision



Record Completion



Product Release

This process ensures that the release decision is supported by appropriate evidence.

4. DEFINITIONS

Product Release

The documented authorization that a completed product has met the applicable requirements and may proceed to the next approved stage.

Non-Conformance

A condition in which a product, process, document or record does not meet a specified requirement.

Traceability

The ability to associate a product with relevant manufacturing, material, inspection and quality records.

Controlled Document

A document maintained under an approved document-management and revision-control process.

Quality Record

Documented evidence that an activity, inspection, review or decision has been completed.

SME

Subject Matter Expert responsible for providing or validating specialist technical information.

Disposition

The approved decision regarding the status or treatment of a product or non-conformance.

5. ROLES AND RESPONSIBILITIES

5.1 Manufacturing Operator

The Manufacturing Operator shall:

- Complete the assigned manufacturing activities.
 - Follow approved manufacturing documentation.
 - Record required production information.
 - Identify abnormal conditions.
 - Submit completed records for review.
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5.2 Production Supervisor

The Production Supervisor shall:

- Confirm manufacturing activities are complete.
 - Ensure required production records are available.
 - Review production-related exceptions.
 - Coordinate with Quality when issues are identified.
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5.3 Quality Inspector

The Quality Inspector shall:

- Perform or verify required inspections.
 - Review inspection results.
 - Confirm applicable acceptance criteria.
 - Identify and document non-conformances.
 - Provide inspection records for product release review.
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5.4 Quality Representative

The Quality Representative shall:

- Conduct the final quality review.
 - Verify required records are complete.
 - Review open quality issues.
 - Confirm applicable acceptance criteria.
 - Approve or reject the product release decision within assigned authority.
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5.5 Engineering

Engineering shall provide technical support when:

- A product requirement requires clarification.
 - A dimensional or functional issue requires engineering assessment.
 - A deviation requires engineering review.
 - A change may affect product fit, form or function.
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5.6 Document Controller / Documentation Owner

The Document Controller or Documentation Owner shall:

- Maintain controlled documentation.
 - Support document revision activities.
 - Ensure current approved documents are available.
 - Maintain applicable document records.
 - Support archival of superseded documentation.
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6. REQUIRED DOCUMENTS AND RECORDS

The following documents and records may be required for product release:

Document / Record	Purpose
Product Specification	Defines product requirements
Engineering Drawing	Defines controlled product geometry
Manufacturing Work Instruction	Defines production activities
Inspection Record	Records inspection results
Material Certificate	Supports material traceability
Non-Conformance Record	Documents product or process issues
Rework Record	Documents approved rework
Engineering Approval	Supports technical disposition where required
Product Release Record	Records final release decision

The latest approved revision of controlled documents shall be used.

7. SYSTEMS AND DOCUMENT MANAGEMENT

Product and supporting documentation may be maintained across approved document-management and enterprise systems.

Relevant systems may include:

Teamcenter

May be used for controlled product-related information, product structures, revisions, datasets and engineering workflows.

Windchill

May be used for product documentation, lifecycle management, revisions and engineering change activities.

SharePoint

May be used for controlled collaboration, document repositories, knowledge sharing and supporting business documentation.

The applicable system of record shall be determined by the organisation's established process.

Users shall not rely on uncontrolled or obsolete copies when performing activities governed by this SOP.

8. PREREQUISITES

Before starting the product quality review, confirm that:

1. Manufacturing activities are complete.
2. The correct product and revision have been identified.
3. Required inspection activities have been completed.
4. Required manufacturing records are available.
5. Applicable material traceability information is available.
6. Relevant non-conformance records have been identified.
7. Required controlled documents are available at their current revision.

If required information is missing, the review shall be placed on hold until the information is obtained or the issue is appropriately resolved.

9. PROCEDURE

9.1 Manufacturing Completion Review

The Production Supervisor shall confirm that the manufacturing process has been completed according to the applicable manufacturing documentation.

Verify:

- Product identification
- Part number
- Product revision
- Production status
- Required manufacturing steps
- Required process records

Any incomplete manufacturing activity shall be addressed before proceeding with product release.

10. DOCUMENTATION COMPLETENESS REVIEW

The Quality Representative shall review the available product documentation.

Confirm that:

- Required records are present.
- Required fields are completed.
- Dates are recorded where required.
- Personnel identification is available where applicable.
- Required approvals or sign-offs are present.
- Records are legible and traceable.
- Applicable document revisions are correct.

Missing or incomplete records shall be returned to the responsible function for correction.

11. INSPECTION RESULT REVIEW

Review the applicable inspection records.

Confirm:

- Required characteristics have been inspected.
- Actual measurements have been recorded.
- Measurements meet the applicable specification.
- Inspection equipment requirements have been satisfied.
- Inspection results are identifiable to the applicable product or batch.

Example:

Characteristic	Requirement	Result	Status
Overall Length	249–251 mm	250.2 mm	PASS
Overall Width	149–151 mm	149.8 mm	PASS
Thickness	24.5–25.5 mm	25.1 mm	PASS
Hole Diameter	21.8–22.2 mm	22.1 mm	PASS
Hole Centre Distance	99.5–100.5 mm	100.2 mm	PASS

The values above are demonstration values only.

12. MATERIAL TRACEABILITY REVIEW

Confirm that the applicable material information is available and traceable.

Verify, where applicable:

- Material grade
- Material certificate
- Heat number
- Batch number
- Supplier identification
- Product identification

Where traceability information is missing or inconsistent, the product shall not be released until the issue is appropriately resolved.

13. NON-CONFORMANCE REVIEW

Review applicable non-conformance records associated with the product.

Determine whether:

- A non-conformance exists.
- The issue has been investigated.
- Required corrective action has been completed.
- Engineering review is required.
- Quality approval is required.
- The product can be released under an approved disposition.

Open issues that could affect product conformity shall be resolved or formally dispositioned before release.

14. REWORK REVIEW

Where rework has been performed, verify that:

1. The rework was authorized.
2. The applicable rework instruction was followed.
3. Required re-inspection was completed.
4. Results meet the applicable requirements.
5. Rework records are complete.

A reworked component shall not be released solely on the basis that the original non-conformance was addressed. Required verification shall also be completed.

15. ENGINEERING REVIEW

Engineering review shall be requested when a technical assessment is required.

Examples include:

- Dimensional deviation
- Product geometry concern
- Material substitution
- Functional concern
- Engineering change impact
- Deviation from approved product requirements

Engineering shall document the applicable technical decision through the approved process.

16. QUALITY RELEASE DECISION

After completing the review, the Quality Representative shall determine the product status.

The product may be assigned one of the following statuses:

RELEASED

All applicable requirements have been satisfied and required documentation is complete.

HOLD

Additional information, inspection, review or action is required before release.

REJECTED

The product does not meet the applicable requirements and cannot be released in its current condition.

CONDITIONAL RELEASE

Where organisational procedures permit, a product may be released under a formally approved condition or restriction.

Any conditional release shall be supported by documented authorization.

17. RELEASE DECISION CRITERIA

The product may be released when all applicable conditions below are satisfied:

- Manufacturing is complete.
- Product identity is confirmed.
- Required inspections are complete.
- Inspection results meet specification.
- Material traceability is available.
- Required documentation is complete.
- Non-conformances are resolved or appropriately dispositioned.
- Required engineering reviews are complete.
- Required approvals are available.
- Product identification is correct.

If any required condition remains unresolved, the product shall remain on hold or be assigned another approved disposition.

18. PRODUCT RELEASE RECORD

The final release decision shall be documented.

Example Product Release Record

Field	Example
Product	WC-100
Part Number	WC-100
Revision	A
Batch Number	WC100-260830-01

Field	Example
Inspection Status	Complete
Traceability Status	Complete
Non-Conformance Status	None Open
Engineering Review	Not Required
Final Decision	RELEASED
Reviewed By	Quality Representative
Review Date	30-Aug-2026

The example values above are fictional.

19. PRODUCT STATUS IDENTIFICATION

Following the final decision, the product status shall be clearly identified according to the applicable site process.

Examples may include:

- Released
- Hold
- Rejected
- Awaiting Inspection

Status identification shall remain traceable to the applicable product and records.

20. RECORD RETENTION

Product quality and release records shall be retained according to applicable organisational requirements.

Records may include:

- Inspection results
- Manufacturing records
- Material certificates
- Release records
- Non-conformance records
- Engineering approvals
- Rework documentation

- Supporting correspondence

Records shall be stored in the approved system or repository.

21. DOCUMENT CONTROL

This SOP is a controlled document.

Users shall confirm that they are accessing the latest approved revision before performing activities governed by this procedure.

Superseded versions shall be removed from points of use or clearly identified as obsolete according to the applicable document-control process.

Changes to this SOP shall be reviewed and approved before release.

22. DOCUMENT CHANGE MANAGEMENT

Changes to the product, manufacturing process, quality requirements or supporting systems may require this SOP to be reviewed.

Examples include:

- Product specification changes
- Manufacturing process changes
- New inspection requirements
- Engineering changes
- Changes to release authority
- Changes to document-management systems
- Changes to traceability requirements

The documentation owner shall assess the impact of significant changes and coordinate revisions with relevant SMEs and process owners.

23. TRAINING REQUIREMENTS

Personnel whose responsibilities are affected by this SOP shall receive appropriate training before independently performing activities governed by the revised procedure.

Training may include:

- SOP overview
- Roles and responsibilities
- Product release workflow
- Documentation requirements
- Quality review requirements
- Non-conformance handling
- Document-control requirements
- System usage

Training completion shall be recorded in the applicable learning or training-management system.

24. ESCALATION PROCESS

Escalation is required when:

- Required documentation is unavailable.
- Product identity cannot be confirmed.
- Inspection results are outside specification.
- Traceability information is incomplete.
- A significant non-conformance is identified.
- Engineering clarification is required.
- A release decision cannot be made using available evidence.

The issue shall be escalated to the appropriate supervisor, Quality representative or Engineering function according to the applicable organisational process.

25. CONTINUOUS IMPROVEMENT

Feedback from manufacturing, quality, engineering and documentation stakeholders should be reviewed periodically to identify opportunities to improve the release process.

Potential improvement areas include:

- Reducing documentation delays
- Improving record completeness
- Eliminating duplicate data entry
- Improving document accessibility
- Clarifying approval responsibilities
- Improving traceability
- Standardizing documentation templates
- Automating repetitive workflow activities

Process improvements shall be implemented through the appropriate change-management process.

26. PROCESS PERFORMANCE INDICATORS

Where applicable, the process owner may monitor:

KPI	Purpose
First-Pass Release Rate	Measures products released without rework or additional review
Documentation Completion Rate	Measures completeness of required records
Release Cycle Time	Measures time from manufacturing completion to release
Open Non-Conformance Count	Tracks unresolved quality issues
Documentation Error Rate	Identifies recurring documentation problems
Rework Rate	Identifies recurring manufacturing or quality issues

These indicators can help identify opportunities for process improvement.

27. RELATED DOCUMENTS

The following documents support this SOP:

- SPEC-WC-100 — Product Specification
 - WI-WC-100-001 — Manufacturing Work Instruction
 - DRG-WC-100 — Engineering Drawing
 - IP-WC-100 — Inspection Plan
 - NC-WC-100 — Non-Conformance Process
 - WC-100 Product Release Record
 - WC-100 Training Guide
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28. DOCUMENT LIFECYCLE

The controlled lifecycle for this SOP is:

Request → Draft → Technical Review → SME Review → Approval → Release → Training → Implementation → Review → Revision → Supersession

Each revision shall be traceable to the applicable change request or documented reason for revision.

29. REVISION HISTORY

Revision	Date	Description	Author
A	30-Aug-2026	Initial demonstration release	Technical Documentation

30. PORTFOLIO DEMONSTRATION NOTICE

This Standard Operating Procedure is an independently created technical writing demonstration developed for a professional portfolio.

WC-100 Industrial Wear Component is a fictional product.

All product characteristics, processes, measurements, roles, workflows, records and scenarios contained in this document are fictional and have been created solely to demonstrate technical writing, manufacturing documentation, quality-process documentation and document-control practices.

This document does not contain confidential, proprietary or customer information belonging to any employer or third party.

This demonstration should not be interpreted as an approved operational procedure for any real manufacturing organisation.

END OF DOCUMENT

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